



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2017/2018

**FIRST YEAR FIRST SEMESTER EXAMINATIONS FOR THE
DEGREE OF BACHELOR OF SCIENCE IN ACTUARIAL SCIENCE
WITH INFORMATION TECHNOLOGY**

MAIN CAMPUS

MAC 105: MICRO-ECONOMICS

Date: 15th February, 2018

Time: 8.30 - 11.30 am

INSTRUCTIONS:

- Answer question ONE and any other TWO questions



MASENO UNIVERSITY

UNIVERSITY EXAMINATIONS 2017/2018

FIRST YEAR FIRST SEMESTER EXAM FOR THE DEGREE OF BACHELOR OF SCIENCE IN ACTUARIAL SCIENCE

Main Campus

MAC 105: MICRO-ECONOMICS

Instructions: Attempt QUESTION 1 and any other TWO questions.

Question 1 – Compulsory [30 Marks]

- a) Explain briefly the following concepts
 - i) Marginal rate of Technical Substitution
 - ii) Production possibility frontier
 - iii) Extensive and Intensive margins of production
 - iv) Inter-temporal choice
 - v) Quasi-rent

[5 Marks]
- b) Give and explain three axioms of rational choice [3 Marks]
- c) Explain the significance of convex shape of isoquant [1 Mark]
- d) Give and explain five factors affecting change in demand of a product [5 Marks]
- e) Give and explain four aspects of a product which may be changed as a method of increasing market share [4 Marks]
- f) Explain the concept insurance and give and explain three major advantages of insurance scheme [4 Marks]
- g) Explain the significance of concave shape of production possibility frontier [1 Mark]

- h) With the help of a diagram, explain a firm's marginal and average product curves in the short-run in the process of production [5 Marks]
- i) What is meant by the term 'economic rent' earned by any of the factors of production [2 Marks]

Question 2 - Optional [20 Marks]

- a) XYZ derives utility from 3 goods: Music (M), Wine (W) and Cheese (C). His utility function is of the simple linear form:- $Utility = U(M, W, C) = M + 2W + 3C$
- i) Assuming XYZ's consumption of music is fixed at 10, determine the equations for the indifference curves for W and C for $U = 40$ and $U = 70$. Sketch these graphs/curves [2 Marks]
- ii) Show that XYZ's MRS of wine for cheese is constant for all values of W and C in the indifference curves computed in part(i) [1 Mark]
- iii) Suppose XYZ's consumption of music increases to 20. How would this change your answers to parts (i) and (ii)? Explain your results intuitively [2 Marks]
- b) With the help of suitable diagrams, describe the circular flow of income indicating points at which the flow may be influenced [5 Marks]
- c) You are given the following production function:-

$$Q = 600X_1^2X_2^2 - X_1^3X_2^3, \text{ where } X_1 = 10$$
- i) Find units of X_1 which corresponds to maximum average product [2 Marks]
- ii) Find units of X_2 which maximize output [2 Marks]
- iii) Comment on the elasticity of production in the second case [1 Mark]
- d) Give reasons and explain why firms set a limiting price [5 Marks]

Question 3 – Optional [20 Marks]

- a) With the help of suitable diagrams, explain the concepts:-
- i) Income effect [5 Marks]
- ii) Substitution effect [5 Marks]
- iii) Forward integration [2 Marks]
- iv) Sunk-cost [2 Marks]
- b) Consider the following utility functions

- i) Utility = $U(X,Y) = XY$
- ii) Utility = $U(X,Y) = X^2Y^2$
- iii) Utility = $U(X,Y) = \ln X + \ln Y$

Show that each of these functions has a diminishing marginal rate of technical substitution, but that they exhibit constant, increasing and decreasing marginal utility respectively. Comment on the results [6 Marks]

Question 4 – Optional [20 Marks]

- a) Explain what is meant by the term ‘economic rent’ earned by any of the factors of production [1 Mark]
- b) Firms can benefit a great deal from expansion and specialization. Explain :-
 - i) Why it may be necessary to limit the growth of a firm [3 Marks]
 - ii) How this process occurs [3 Marks]
- c) Give and explain the five types of insurance business transacted in most modern economics [5 Marks]
- d) Outline three necessary conditions, if there is to be perfect competition in the market [3 Marks]
- e) Explain the concept advertising intensity [2 Marks]
- f) Explain what is meant by the term ‘price discrimination’. In what circumstances/reasons and for what reasons is a monopolist likely to discriminate in his pricing policy [3 Marks]

Question 5 – Optional [20 Marks]

- a) Explain the firm’s marginal and average cost curves:-
 - i) In the short run [5 Marks]
 - ii) In the long-run in the production [5 Marks]
 (Use diagrams in your explanation)
- b) Rent is a surplus and not a cost and as such it does not determine the price of goods [5 Marks]
- c) Give reasons and explain why firms set up a limiting price [5 Marks]



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2017/2018

**FIRST YEAR FIRST SEMESTER EXAMINATIONS FOR THE
DEGREE OF BACHELOR OF SCIENCE IN ACTUARIAL SCIENCES,
MATHEMATICAL SCIENCE AND APPLIED STATISTICS WITH
INFORMATION TECHNOLOGY**

MAIN CAMPUS

MAC 107: INTRODUCTION TO ACTUARIAL SCIENCES

Date: 14th February, 2018

Time: 3.30 - 6.30 pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions
- Show all the necessary workings



QUESTION ONE

- a) Briefly explain the seven principles of insurance, giving an example in each case (not examples given in class). [7 marks]
- b) Describe the following types of insurances;
- Term insurance [2 marks]
 - Endowment insurance [2 marks]
 - Pure endowment. [2 marks]
- c) Suppose you agree to deposit sh. 500 at the end of every year for three years in an investment fund that earns 6%. How much will you earn at the end of the three years? [5 marks]
- d) At the end of each month, a saver deposited sh. 100 into a savings account that paid 6% compounded monthly. How much would he have at the end of 10 years? [5 marks]
- e) In a certain country, censuses are held on 30th june each year. The enumerated male population on 30th june, 1993, 1994 and 1995 at ages 30, 31, 32 and 33 last birthday was as follows.

Age	Year		
	1993	1994	1995
30	306421	303606	307412
31	300112	305169	302121
32	299408	299168	304003
33	299133	298977	299015

Estimate the central exposed to risk from 1993 to 1995 at each ages 30, 31, 32 and 33 last birthday using

- The repeated mid-point rule
- The formula used in ELT No. 14. [6 marks]

QUESTION TWO

- a) A customer is offered an investment where interest is calculated according to the following force of interest

$$\delta_t = \begin{cases} 0.02t & 0 \leq t \leq 3 \\ 0.045 & t > 3 \end{cases}$$

The customer invests sh. 1000 at time $t=0$. What nominal rate of interest compounded quarterly is earned over the first four year period? [4 marks]

- b) Jane , Lori and Lucy each borrow a loan of sh. 5000 for five years at a nominal interest rate of 12% compounded semi-annually.
Jane has interest accumulated over the five years and pays all the interest and principal in a lump sum at the end of five years.
Lori pays interest at the end of every six-month period as it accrues and the principal at the end of the five years.
Lucy repays her loan with 10 level payments at the end of every six-month period.
Calculate the amount of interest paid on the three loans. [8 marks]
- c) Compare and contrast the graphical method and mathematical formula of graduation. [8 marks]

QUESTION THREE

- a) Discuss the main types of life insurance. [8 marks]
- b) Bruce and Robbie each open up new bank accounts at time 0. Bruce deposits sh. 100 into his bank account and Robbie deposits sh. 50 into his. The amount of interest earned in Bruce's account during the 11th year is equal to x . The amount of interest earned in Robbie's account during the 17th year is also equal to x . Calculate x . [5 marks]
- c) Suppose you deposit sh. 100 at the end of year one, 200 at the end of year two and 300 at the end of year three. Find the present value if the annual interest rate is 10%. [3 marks]
- d) Differentiate between the nominal rate of interest and the effective rate of interest. [4 marks]

QUESTION FOUR

- a) Suppose you are offered an investment that will pay you sh. 200000 in year one, sh. 400000 in year two, sh. 600000 in year three and sh. 800000 in the

- final year. In this investment you earn an interest rate of 12%. What is the most you should pay for? [5 marks]
- b) You can buy a security now for sh.1000 and it will pay you sh. 1191 three years from now. What annual rate of return are you earning? [3 marks]
- c) Given that a savings bank charges a nominal interest rate of 10% compounded monthly. What is the corresponding effective annual rate of interest? [2 marks]
- d) Suppose that $\mu_{x+0.5}(\alpha) = \alpha \cdot \mu_{x+0.5}^T$. Find a simple formula for $\hat{\alpha}$, the Maximum Likelihood Estimator of the true value of $\alpha_1 = \alpha$. [4 marks]
- e) Discuss the three methods of investment appraisal techniques. [6 marks]

QUESTION FIVE

- a) Show that equivalent nominal interest and discount rates i^m and d^m satisfy the relationship $d^m = \frac{i^m}{1 + \frac{i^m}{m}}$. [5 marks]
- b) Discuss any five types of insurance products offered by insurance companies. [10 marks]
- c) Let P_x^0 and P_x^1 denote the number of lives present and aged x last birthday at the beginning and end respectively of a year-long investigation and let θ_x denote the number of deaths at age x last birthday during the year. Show that $E_x = \frac{1}{2} [P_x^0 + P_x^1 + \theta_x]$. [5 marks]